

Technology Stack

Date	28 June 2026
Team ID	
Project Name	Smart Lender
Maximum Marks	2 Marks

Define Your Technology Stack:

Identify the programming languages, frameworks, databases, front-end tools, back-end tools, and APIs that you will use to build your solution. A well-chosen technology stack ensures that your application is scalable, maintainable, and aligned with your project requirements.

Fill out the table below to detail the specific technologies chosen for each component of your architecture and provide a brief justification for your choice.

Technology Stack Details

S.No	Architecture Component / Layer	Technology Chosen	Justification / Purpose
1	Frontend / Client-Side	HTML5, CSS3, JavaScript	Provides a responsive and user-friendly interface for entering loan details and displaying prediction results.

2	Backend / Server-Side)	Python, FastAPI	FastAPI provides high-performance API development, efficient request handling, and seamless integration with the machine learning model.
3	Database / Data Storage	CSV Dataset, model.pkl, scaler.pkl	Stores the loan dataset for training and the serialized machine learning model and scaler for prediction.
4	Cloud / Hosting / Deployment		Hosts the web application securely, enabling browser-based access, scalability, and easy deployment.
5	Version Control & CI/CD	Github	Supports version control, collaboration among team members, source code management, and project tracking.

6	Third-Party APIs / Other Tools	Scikit-learn, XGBoost, Pandas, NumPy, Matplotlib, Seaborn	Used for data preprocessing, exploratory data analysis, model training, evaluation, visualization, and loan eligibility prediction.
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